Pellevoisin: the Raspberry Pi 4 header

Pin 1 top left with the board's edge at the top: odd pins in the inner row, even pins in the outer row next to the edge. Highlighted cells are this organ's connections. Every GND pin is the Pi's ground, and nothing on the organ's side ever touches this header.

ours	function	pin	pin	function	ours
MIDI: 6N137 module IN+	**3.3 V**	**1**	**2**	5 V	free (Pi fed by USB-C from the 5 V supply in the box)
free; I2C stays off	GPIO 2 / SDA	**3**	**4**	5 V	free
ON/OFF SWITCH, one side	**GPIO 3 / SCL**	**5**	**6**	**GND**	**ON/OFF SWITCH, other side**
	GPIO 4	**7**	**8**	**GPIO 14 / TXD**	**MIDI: 6N137 module IN-**
RESET opto cathode (PC817 pin 2)	**GND**	**9**	**10**	GPIO 15 / RXD	free
RESET: 330 ohm to PC817 anode	**GPIO 17**	**11**	**12**	GPIO 18	
STATUS LED: 330 ohm to LED anode	**GPIO 27**	**13**	**14**	**GND**	**STATUS LED cathode**
	GPIO 22	**15**	**16**	GPIO 23	
	3.3 V	**17**	**18**	**GPIO 24**	**PUMP: RJ45 3, 100R to MOSFET gate in box**
	GPIO 10	**19**	**20**	**GND**	**PUMP: RJ45 6, MOSFET source**
	GPIO 9	**21**	**22**	GPIO 25	
	GPIO 11	**23**	**24**	GPIO 8	
	GND	**25**	**26**	GPIO 7	
never	ID_SD	**27**	**28**	ID_SC	never
	GPIO 5	**29**	**30**	GND	
	GPIO 6	**31**	**32**	GPIO 12	
	GPIO 13	**33**	**34**	GND	
	GPIO 19	**35**	**36**	GPIO 16	
	GPIO 26	**37**	**38**	GPIO 20	
	GND	**39**	**40**	GPIO 21	

Each line is paired with its nearest ground so every cable is a short twisted pair: 5 and 6 for the switch, 11 and 9 for reset, 13 and 14 for the status LED, 18 and 20 for the pump. The MIDI pair is pins 1 and 8 and carries no ground: TX is its return. Pins 3 and 5 are the I2C bus, which stays disabled so GPIO 3 is free for the switch and wakes a halted Pi when pulled low. The DSI screen uses its own ribbon connector.

Part pinouts: PC817 pin 1 anode, 2 cathode, 3 emitter, 4 collector (dot at pin 1). SSR-xx DA: 1 and 2 load (mains), 3 positive and 4 negative input, 3 to 32 V DC. 6N137 module: IN+ and IN- marked on the terminals; VCC, OUT, GND on the other side. Pi 5 V: a Mean Well 5 V mains supply in the box, 3 A fuse, over the Ethernet cable's three spare pairs (pins 1,4,7 plus; 2,5,8 return; 3,6 the SSR), trimmed to 5.1 V at the Pi, USB-C pigtail. organ_web: --pump 24 --power-switch 3; reset on 17. Status LED: config.txt dtoverlay=gpio-led,gpio=27,label=pellevoisin,trigger=heartbeat; it beats while Linux runs and is dark when halted. Enclosure fan: 5 V 4-wire, always on, +5 V and GND from the terminal where the RJ45 5 V pairs feed the USB-C pigtail (never header pins 2/4); tach and PWM wires insulated, unconnected PWM = full speed. Pump SSR: box 5 V to SSR 3, SSR 4 to a logic-level MOSFET's drain (VGS(th) <= 2 V), source to RJ45 6, gate 100R from RJ45 3 and 100k to source; never 12 V on the SSR (grounds).

Pellevoisin: the box, wired

Two halves in one aluminium case, and exactly three things cross between them, each by light: the MIDI opto, the reset opto and the SSR. The Pi has a mains supply of its own, in the box, and its 5 V travels in the Ethernet cable beside the SSR pair.

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MAINS -- inlet, 15 A breaker -+- E ---- earth stud on the case ---- motor frame
                              +- N ---------------+------------------------------- motor N
                              |                   PSU N
                              +- L -- PANEL SWITCH, 2 poles, 3 positions, centre off, motor rated
                                       pole A common -+- BOOK ----------------------> motor L
                                                      +- AUTO -> SSR 1 -- SSR 2 ------> motor L
                                       pole B common --- AUTO -> 12 V PSU L (green on wire to GND) and 5 V PSU L
                                       (OFF: neither pole connected; the supply's fan cools the box in Auto)

ORGAN SIDE (the 12 V supply's ground)                PI SIDE (the 5 V supply's ground)
PSU 12 V, several wires --> 12 V bus bar             5 V PSU (mains, in the box) - RJ45 pairs 1-2,4-5,7-8 -USB-C-> P
PSU GND, several wires ---> organ GND bus bar        DSI ribbon --> touchscreen;  same 5 V terminal --> enclosure fa
PSU on wire (green) ------> PSU GND: on with Auto     GPIO 3 (pin 5) -- antique switch -- Pi GND (pin 6)   on/off
12 V bus / GND bus -------> boards 1-4, 5 A fuses    GPIO 14 TX (pin 8) -- 6N137 IN-             MIDI
                                                     3.3 V (pin 1) ------- 6N137 IN+
                                                     GPIO 24 (pin 18) -- RJ45 pin 3 - MOSFET gate pump
last board ISP pin 2 -----> RJ12 pin 6 = 5 V bus     Pi GND (pin 20) --- RJ45 pin 6 -- MOSFET source; SSR 3 < box 5
RJ12 pin 1 = /RESET bus (jumpered board to board)    GPIO 27 (pin 13) -[330 ohm]- LED - Pi GND (pin 14)  alive
RJ12: MIDI signal, GND, as shipped                   GPIO 17 (pin 11) -[330 ohm]- PC817 anode         reset
6N137 module: VCC < RJ12 pin 6, GND < RJ12 GND,      Pi GND (pin 9) ------------ PC817 cathode
              OUT > boards' MIDI in
PC817: collector - RJ12 pin 1, emitter - RJ12 GND
SSR 1 / 2: on the mains side, above; MOV across them
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Rules

- No wire, bar or chassis lug between the organ's GND bus and the Pi's ground: the Pi's 5 V comes from its own mains supply, never from a buck on the 12 V bus. The earth stud carries earth only, and no DC negative goes to it.
- Mains bus bars shrouded, or covered DIN terminal blocks instead. Low-voltage bars may be bare.
- Mains and low voltage on opposite sides of a partition: inlet, switch, supplies' inputs, SSR 1 and 2, motor lead on one side; Pi, RJ12 bus, optos, SSR 3 and 4 on the other. A wire that must cross does so once, at right angles.
- The SSR's heat sink in the 12 V supply's fan stream, thermal paste under the module. A varistor across SSR 1 and 2.
- Only the live is switched, ever. Neutral and earth run straight through. The panel switch is motor rated: 1 HP at 125 V, or 20 A. The antique on/off switch carries microamps and may be anything.
- Book: the motor runs, nothing else has power. Auto: the 12 V supply (Lenovo SP50H29523, 15 A) starts at once, its green on wire tied to its ground, and the bus, the fan, the boards and the Pi on its own 5 V supply come up together; the motor runs only when the SSR says so. Off: everything isolated (the SSR alone leaks a milliamp).
- Before the panel goes from Auto to Off: the antique switch off, or Shut down on the Service tab, and wait for the status LED to go dark.